

Classification of Obsessive Compulsive Disorder by EEG Complexity and Hemispheric Dependency Measurements

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In the present study, both single channel electroencephalography (EEG) complexity and two channel interhemispheric dependency measurements have newly been examined for classification of patients with obsessive-compulsive disorder (OCD) and controls by using support vector machine classifiers. Three embedding entropy measurements (approximate entropy, sample entropy, permutation entropy (PermEn)) are used to estimate single channel EEG complexity for 19-channel eyes closed cortical measurements. Mean coherence and mutual information are examined to measure the level of interhemispheric dependency in frequency and statistical domain, respectively for eight distinct electrode pairs placed on the scalp with respect to the international 10–20 electrode placement system. All methods are applied to short EEG segments of 2 s. The classification performance is measured 20 times with different 2-fold cross-validation data for both single channel complexity features (19 features) and interhemispheric dependency features (eight features). The highest classification accuracy of $85 \pm 5.2\%$ is provided by PermEn at prefrontal regions of the brain. Even if the classification success do not provided by other methods as high as PermEn, the clear differences between patients and controls at prefrontal regions can also be obtained by using other methods except coherence. In conclusion, OCD, defined as illness of orbitofronto-striatal structures [Beucke *et al.*, *JAMA Psychiatry* **70** (2013) 619–629; Cavedini *et al.*, *Psychiatry Res.* **78** (1998) 21–28; Menzies *et al.*, *Neurosci. Biobehav. Rev.* **32**(3) (2008) 525–549], is caused by functional abnormalities in the pre-frontal regions. Particularly, patients are characterized by lower EEG complexity at both pre-frontal regions and right fronto-temporal locations. Our results are compatible with imaging studies that define OCD as a sub group of anxiety disorders exhibited a

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decreased complexity (such as anorexia nervosa [Toth et al., *Int. J. Psychophysiol.* **51**(3) (2004) 253–260] and panic disorder [Bob et al., *Physiol. Res.* **55** (2006) S113–S119]).

Keywords: Approximate entropy; sample entropy; permutation entropy; mutual information; SVM.

1. Introduction

Obsessive–compulsive disorder (OCD) is a mental disorder characterized by the presence of obsessions or compulsions or both. Obsessions are unwanted and intrusive thoughts that are recurrent and persistent whereas compulsions are repetitive behaviors, mental acts or rigidly applied rituals that the patients feel driven to perform in response to obsessions.¹ OCD was modeled by a chaotic neural network to assess the treatment in computational manner.² In several studies, spectral estimations^{3–5} and independent component analysis^{6,7} were applied to particular frequency bands of spontaneous electroencephalography (EEG) series recorded from relaxed volunteers to assess a specified treatment. In previous studies, most of the intelligent diagnostic methods consisting of EEG classification have been developed for seizure detection.^{8–11,79} In all of them, seizure activity and epilepsy disorder were commonly identified based on the quantities in terms of EEG complexity. To our knowledge, we have firstly investigated the degree of EEG complexity for OCD classification in the present study. In addition, we have newly introduced regional interhemispheric dependency metrics to discriminate OCD from controls.

Entropy estimation is the most common approach in measuring the degree of EEG complexity due to nonstationary nature of the EEG series.^{9,12–14} In past, Kolmogorov–Sinai (KS) and Eckmann and Ruelle entropies were developed to measure the regularity of random series, however, these approaches were found to be not well suited for characterization of short time noisy signals.^{24,25} Therefore, the approximate entropy (ApEn) was proposed as a modification of KS entropy of short time series.¹⁵ As a result of wide range of applications, capability of ApEn was found to be sensitive to both spectral bandwidth and amplitude distribution in characterizing of short noisy signals.¹⁶ In performing ApEn, several samples might be compared to itself when two consecutive sliding time windows commonly consist of the same samples. So, this self-similarity is

also counted wrongly. Another disadvantage of ApEn is that its performance highly depends on the data length.^{12,17} Sample Entropy (SampEn) was proposed to overcome those handicaps of ApEn.¹² SampEn has been frequently used to characterize the spontaneous EEG signals for not only detection of seizure¹⁸ but also discrimination of alcoholics from controls.¹⁰ SampEn was also examined for observation of the effect of anesthesia on younger and elderly adults.¹⁹ Both ApEn and SampEn require *a priori* selection of two unknown parameters such as embedding dimension depending on the data length and the similarity criterion so-called threshold depending on the standard deviation of the data.^{10,12,17–19}

Permutation Entropy (PermEn) is recently introduced as embedding like entropy estimation method to assess the complexity of chaotic signals such as EEG series. PermEn provides information about how time series fluctuates with time in an ordinal way. PermEn can also be considered as an invariant parameter, which is able to acquire not only slowing effect but also loss of complexity.²⁰ Therefore, PermEn was found to be capable of recognizing the differences between irregular and regular behavior of cerebral electrical activity, to track transient dynamics of EEG recordings²¹ and to detect seizure.^{22,23} Different from both ApEn and SampEn, PermEn transforms the time series on to a symbolic space, and then it measures the regularity of this symbolic series. In the present study, both ApEn and SampEn as well as PermEn have newly been examined for discrimination of OCD from controls by means of single channel EEG complexity. In fact, those entropy estimation approaches are widely used to both analyze nonlinear systems and characterize the degree of randomness of time sequences in information theory. Fractal dimension (FD) was also used to investigate the complexity of EEG frequency bands in children with autistic spectrum disorder⁶⁹ and elders with Alzheimer's disease⁷⁰ by means of chaos theory. Different from OCD, both autism and Alzheimer are progressive brain disorders consist of a number of

common symptoms such as difficulty with language comprehension. A wavelet chaos methodology was introduced for analysis of EEG in detecting seizure that is defined as neuronal firing represented by less organized EEG signal with greater chaoticity due to the fact that EEG is superimposition of diverse processes in the brain.^{80,81} In those studies, EEG sub-bands (delta (0–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz) and gamma (30–60 Hz)) have been commonly obtained by using wavelet decomposition. This wavelet-based chaos theory was also adopted to develop new diagnostic markers in not only Alzheimer’s disease^{82,83} but also ADHD^{84,85} as well as Autism spectrum disorder.^{86,87}

It is possible to measure both linear and non-linear dependencies of two random series by computing their mutual information (MI) in statistical domain.²⁶ Due to stochastic behavior of EEG series, MI has been largely studied to compute the inter-hemispheric dependency with respect to two channel EEG measurements coming from the electrodes symmetrically placed on the scalp.²⁷ MI was reported as useful for detection of psycho-physiological states of individuals in schizophrenia²⁸ and Alzheimer.^{29,71} Interhemispheric correlations can also be measured

by using frequency domain spectral methods such as coherence.^{30,31} However, coherence was found to be reference dependent,^{31,32} and task related³² in several clinical studies due to the fact that coherence can measure only linear dependence between random series. Thus, both MI and coherence were used to estimate the EEG synchronization for each couple of electrodes for 16-channel eyes closed surface EEG recordings to classify the patients with OCD and controls in the present study.

The experimental data, consisting of 19 channels eyes closed EEG series, were collected from drug-naïve patients and controls. Since, the previous studies reported that brain activities could be changed by pharmacotherapy,^{33,34} the patient profile of drug-naïve is important. Due to nonstationary nature of EEG signals, eyes closed cortical measurements were divided into short EEG segments of 2 s before analysis.^{23,35} Then, three embedding entropy approaches (ApEn, SampEn, PermEn), and two channel regional interhemispheric dependency estimation (MI, coherence) were examined for each short EEG segment within the second measurement interval of 20 s. Then, support vector machines (SVM) were used to classify the patients with OCD and

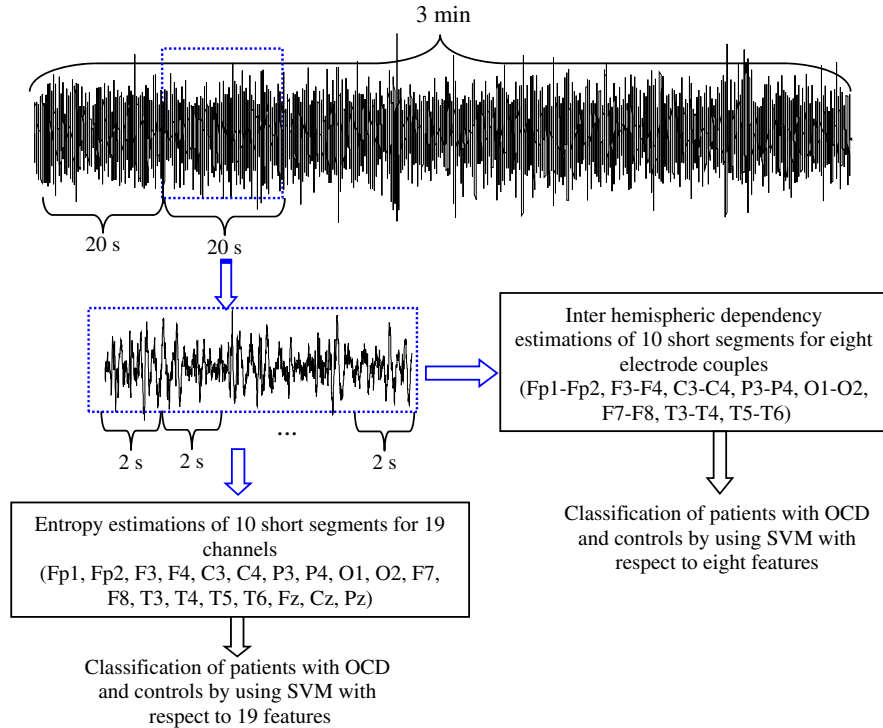


Fig. 1. Block diagram of the applications in the present study.

controls with respect to estimated features as graphically shown in Fig. 1. Given the training samples from OCD and control categories, the parameters of optimal separating hyperplane in feature space were found iteratively by gradient descent algorithm. In test stage, those EEG features were simply classified based on their distance to the separating hyperplane. The experimental results shows that OCD is characterized by lower EEG complexity and PermEn provides the highest classification accuracy at prefrontal locations.

The paper is organized as follows: We first explain the data collection procedure in Sec. 2. Then the feature extraction and classification methods are explained in Sec. 3. After the experimental results given in Sec. 4, discussion and conclusions are given finally.

2. Data Collection

All patients presenting to Uskudar University Neuropsychiatry Health Practice and Research Center who had never received psychiatric treatment were assessed with EEG in addition to other standard tools of neuropsychiatric evaluation. Patients, having demographic information as given in Table 1, were diagnosed according to the DSM-4.³⁶ Exclusion criteria included a history of traumatic head injury, schizophrenia or any other psychotic disorder, bipolar I or II disorder, alcohol and substance abuse, left-handedness, mental retardation or any neurological disorder such as epilepsy. EEG data were also acquired for age- and sex-matched healthy

Table 1. The demographic and clinical characteristics of the patients. (F: female, M: male, O: obsession, C: compulsion, s: score, $d1$ and $d2$ refer the duration from the appearance of initial symptoms in years and obvious symptoms in months, respectively.)

Sex	Age	Y-BOCs	os	cs	H-17	BAI	$d1$	$d2$
F	25	26	12	14	13	32	17	48
M	26	23	11	12	5	5	2	12
M	23	23	11	12	14	11	5	24
F	33	22	8	14	6	28	15	72
M	28	21	10	11	10	10	22	12
M	29	16	14	2	8	11	11	36
F	22	15	13	2	19	18	11	8
F	31	15	9	6	20	18	5	12
M	42	14	6	8	9	11	22	24
F	48	12	12	0	19	24	7	6

controls. The university Ethics Committee approved the research protocol, and voluntary participants signed a form of written consent. We obeyed the guidelines of the Helsinki Declaration. As universal clinical applications, the severity of OCD was assessed with respect to individual's answers to 10 of 19 questions in the Yale-Brown Obsessive Compulsive Scale (Y-BOCs) severity scale with the range of 0–40.³⁷ In addition, the 17 item Hamilton Depression Rating Scale (H-17)³⁸ was used to measure the severity of depression where the total score of which can be in the range 0–51. The anxiety score was estimated by employing the Beck Anxiety Inventory,³⁹ a 21-item self-report questionnaire, the total scores of which can be in the range 0–63. Our patients were all drug naive. Scalp EEG series were recorded from subjects using a 19-channel Neuroscan Synamps II (Neuroscan Products, Compumedics, USA) recording system. Ag/AgCl surface electrodes were attached to the scalp surface of volunteers in accordance with the international 10–20 recording system using a quick cap within a light controlled recording room. During measurements, the temperature was set to approximately 22°C. The sampling frequency was 250 Hz. 12-bit analog-to-digital converter was used in measurements. The impedance values of electrodes were kept less than 5 k Ω . The rejection level was 50 μ V peak-to-peak for EOG signals to remove artifacts created by eye blinks or movements. An electrode was placed inferior to the right eye, and two electrodes were located on the left and right outer canthi of the eyes to measure both vertical and horizontal bipolar EOG signals. A low pass filter (0.5–70 Hz) was applied to raw data in Scan Edit 4.3 software. A Notch filter was applied to suppress the 50 Hz power-line interference. The muscle artifacts and physiological artifacts such as eye blink were eliminated by a technician, who carefully inspected the recordings.

3. Methods

In the present study, we followed two concepts to characterize the eyes closed surface EEG series collected from controls and drug naive patients with OCD as follows: Firstly, three embedding entropy methods so called ApEn, SampEn and PermEn were examined to obtain single channel EEG complexity estimations. Secondly, MI and coherence

were used to obtain two channel regional interhemispheric dependency estimations. In many experimental studies, it was shown that EEG series is stationary for just a short EEG segment of 2s.^{23,35,40–43} Accordingly, each single trial EEG measurement was segmented by using a constant window of 2s before applying all those methods as shown in Fig. 1.

3.1. Single channel EEG complexity estimation

Entropy is conceptually defined as the uncertainty of a random variable in information theory.¹³ Entropy can measure the level of irregularity/complexity of finite time series such that an ordered signal produces the lower entropy.^{9,10} Theoretically, entropy quantifies the randomness of one-dimensional finite length time series.^{9,10} If the probability of occurrence of each segment in time series does not change throughout the recording interval, the high entropy has to be computed. However, the algorithm needs arrangement of crucial parameters in calculating the entropy.

The randomness of stochastic signals can be measured by using ApEn with respect to logarithmic likelihood in which sets of patterns are close on next incremental comparison.¹⁵ ApEn is a statistical tool, which was introduced as a result of the modification of KS which can be defined as the mean rate of information generated by a random process.^{65,66} In the field of nonlinear dynamics, it was found to be useful for characterization of short time signals in measuring the logarithmic likelihood that runs of patterns that are embedding vectors in form,

$$x_m(i) = [x(i), x(i+1), \dots, x(i+m-1)]. \quad (1)$$

Here, $i = 1, 2, \dots, N - m + 1$ and m denotes the run length, namely the embedding dimension where the likelihood that m nearby points on the trajectory remain close to each other. Thus, ApEn is in fact the average conditional information created by diverging points on the trajectory. The distance between two embedding vectors is formulized by,

$$d = \max_{k=1,2,\dots,m} \|x(i+k-1) - x(j+k-1)\|. \quad (2)$$

The correlation sum of embedding vectors is defined by,

$$C_i^m(r) = \frac{\# \text{ of } d[x_m(i), x_m(j)] \leq r}{N - m + 1} \quad (3)$$

within a tolerance level denoted by r for $i = N - m + 1$. Taking the natural logarithm of each $C_i^m(r)$

and, averaging it over i , ApEn is calculated by the following formula,

$$\begin{aligned} \text{ApEn}(m, r, N) = & \frac{1}{(N - m + 1)} \sum_{i=1}^{N-m+1} \ln C_i^m(r) \\ & - \frac{1}{(N - (m + 1) + 1)} \\ & \times \sum_{i=1}^{N-(m+1)+1} \ln C_i^{m+1}(r). \end{aligned} \quad (4)$$

In computing the value of entropy by using the algorithm of ApEn, several samples might be compared to itself since any sample might be commonly placed in two consecutive sliding windows and this self-similarity is also counted wrongly as a result. Therefore, SampEn have been developed as a modified version of ApEn to overcome this bias.¹² There are both similarities and diversities between ApEn and SampEn from a methodological point of view. SampEn provides the negative logarithm of the conditional probability in contrast to ApEn. Regarding as Eq. (1), by defining $A_i^m(r)$ and $B_i^m(r)$ for each i where $i \neq j$ as follows,

$$A_i^r(r) = \frac{1}{N - m - 1} (\# \text{ of } d[x_m(i), x_m(j)] \leq r), \quad (5)$$

$$B_i^r(r) = \frac{1}{N - m - 1} (\# \text{ of } d[x_{m+1}(i), x_{m+1}(j)] \leq r). \quad (6)$$

SampEn is calculated in form,

$$\text{SampEn}(m, r, N) = -\ln \left(\frac{\frac{1}{N-m} \sum_{i=1}^{N-m} A_i^m(r)}{\frac{1}{N-m} \sum_{i=1}^{N-m} B_i^m(r)} \right), \quad (7)$$

where $1 \leq j \leq N - m (j \neq i)$ as introduced in Ref. 11. Due to the nature of negative logarithm, a less random (more complex) time series generates the higher SampEn values. As suggested by scientific works,³⁵ ApEn was applied to very short EEG series of 2s in the present study.

Different from SampEn, the PermEn was proposed to sort the values of each segment (expressed in Eq. (1)) in an ascending order for quantification of the relative occurrence of each sample in EEG series considered as a sequence of ordinal patterns with no prior assumption.²¹ In fact, the PermEn is a symbolic analysis which transforms the original

time series onto a symbolic sequence to compare the order of neighboring relative values instead of amplitudes. Algorithmically, the embedded segments of a time series with N samples ($u(1)u(2), \dots, u(N)$) is defined by,

$$X_m(i) = [u(i), u(i+L), \dots, u(i+(m-1)L)], \quad (8)$$

where m and L denote the embedding dimension and the time delay, respectively for $i = 1, 2, \dots$.^{9,10} In Ref. 21, L is set to 1. So, any vector $X_m(i)$ is uniquely mapped onto $(j_1, j_2, \dots, j_m)$ by arranging samples in an increasing order in form $[u(i+(j_1-1)L) \leq u(i+(j_2-1)L) \leq \dots \leq u(i+(j_m-1)L)]$. When all the components of a vector have the same value, $X_0(i) = [u(0), u(0), \dots, u(0)]$ is mapped onto the $[1, 2, \dots, m]$. Resulting from this process, each sample of the m -dimensional embedded vector is mapped onto one of the $m!$ permutations. In computing the PermEn, m -dimensional embedded vectors can be represented by a symbol sequence, since each permutation is identified by a symbol.²¹ If the probability distribution of K distinct symbols are denoted by $P_1, P_2, \dots, P_K$ where $K \leq m!$, the normalized entropy value so-called PermEn is calculated in form,

$$\text{PermEn}(m) = -\sum_{j=1}^K P_j \ln P_j / \ln(m!). \quad (9)$$

In general, the more complex time series generates the higher PermEn.²¹

3.2. Regional two channel interhemispheric dependency estimation

The MI can measure cross dependency between two time series having N samples such as $X = \{x_1, x_2, \dots, x_N\}$ and $Y = \{y_1, y_2, \dots, y_N\}$ with probability distribution of amplitudes given by $p(x_i)$ and $p(y_i)$, respectively as follows,

$$\text{MI}(X, Y) = H(X) + H(Y) - H(XY), \quad (10)$$

where $H(X)$ and $H(Y)$ denote the average amount of information gained from measurements of X and Y , respectively regarding as Shannon information theory in form,

$$H(X) = -\sum_{i=1}^N p(x_i) \log(p(x_i)), \quad (11)$$

$$H(Y) = -\sum_{i=1}^N p(y_i) \log(p(y_i)). \quad (12)$$

The MI gives the information for Y knowing X or vice versa. In other words, MI is a symmetric measurement. In the present study, MI is used to compute interhemispheric dependency between two channel EEG series such as Fp1-Fp2, F3-F4, O1-O2, F7-F8, T3-T4, T5-T6. For example, if the variable X represents left prefrontal recordings (Fp1) of 2 s, with $N = 500$, Y represents right prefrontal recordings (Fp2) of 2 s, with $N = 500$, then if the entropy of X is zero, each possible measurement occurs with a probability of either 1 or 0. In detail, MI can never be larger than any of the individual entropies such that $\text{MI}(X, Y) \leq \min\{H(X), H(Y)\}$.

The advantage of MI is that both linear and non-linear dependency between X and Y are measured by using MI. It is also possible to quantify the linear dependency between X and Y by using a cross spectral analysis so-called coherence in form,

$$\text{Coherence}(f) = \frac{|P_{XY}(f)|^2}{\sqrt{P_X(f)P_Y(f)}}, \quad (13)$$

where f denotes the frequency. The coherence can be computed in the range of 0 to 1 as the ratio of auto-spectra and cross-spectra. The coherence will be equal to 0 when X and Y are not correlated with constant phase shifts. In other words, the coherence becomes 1, if there is a maximum linear interdependence between X and Y .⁴⁴ In the present study, because of its superior performance on noisy signal, Welch's averaged, modified periodogram method was performed in computing interhemispheric coherence as stated in Ref. 45.

3.3. EEG classification

The proposed EEG features were classified using SVM, that has been recently used successfully in many biomedical diagnostics^{46-48,77} and other classification applications.⁷³

For automated detection of disorders, several learning algorithms such as action perception cycle learning⁸⁸ and stacked sequential learning⁸⁹ have been used to provide useful computer-based diagnostic tools in clinical use. The SVM is a discriminative model-based classification technique, which separates the samples of different categories by a discriminant function. Given a set of training samples labeled as positive or negative, SVM builds a model

that assigns unknown features into one of two categories. It basically finds an optimal separating hyperplane between the samples of two categories by maximizing the margins between the closest data samples with different label. The parameters of hyperplane can be written in terms of some samples in the training set. Those samples are called support vectors and are used in calculation of the output of discriminant function.

Although SVM is mainly designed for two-class separation problem, one may transform it into multi-class classification with one-versus-all method.⁴⁹ In addition, we can achieve nonlinear discriminant functions via SVM by transforming the input features into a higher dimensional feature space. For this purpose, instead of applying a nonlinear SVM function, one may use nonlinear kernel functions to get more complex input features so as to increase the classification performance. In recent EEG classification applications, researchers have used several different learning algorithms such as least squares classifiers,⁹⁰ adaptive background modeling⁹¹ and local decision circles⁹² to select the optimum features automatically to obtain high classification performance.

Given the training samples from OCD and control categories, each represented with the proposed EEG features, the parameters of optimal separating hyperplane is found iteratively by gradient descent algorithm. As kernel function, we apply Gaussian Kernel Function with user defined sigma value. Another user defined parameter, in addition to sigma value in the SVM formulation, is the constant C , that trades off between max-margin objective and the L2 parameter regularization in the optimization formulation. In training stage these user defined parameters are determined using cross-validation. Simply, we split the train data into many batches randomly and change the parameters in pre-defined intervals (i.e. 0.01–0.3 for sigma and 1–500 for C). We use 2-fold cross-validation to get the optimal parameters while we do grid-search on sigma and C constant. After training stage, the EEG features used as test samples are simply classified based on their distance to the separating hyperplane). We built a binary SVM classifier which distinguishes one of the labels and the rest (*one-versus-all*). Since there are only two classes in our study, namely patients and controls, we classify a test sample based on which side of the separating hyperplane it falls on.

4. Results

In the present study, three different embedding entropy approaches were used to compute single channel EEG complexity values. In addition, MI and coherence were implemented to compute two channel regional interhemispheric dependency levels in statistical domain and frequency domain, respectively. Those methods were commonly applied to short EEG segments of 2 s within the recording interval of 20 s and then, we had a collection of 90 samples from each group for 19 EEG single channels (Fp1, F3, C3, P3, O1, F7, T3, T5, Fp2, F4, C4, P4, O2, F8, T4, T6, Fz, Cz, Pz) and eight couple of EEG channels (Fp1-Fp2, F3-F4, C3-C4, P3-P4, O1-O2, F7-F8, T3-T4, T5-T6) as shown in Fig. 1. In computing entropy values, data length denoted by N was equal to 500, the embedding dimension denoted by m was set to 4 and tolerance level denoted by r was computed as 2.5 times std of short EEG segment of 2 s for both ApEn and SampEn as well as PermEn. The optimum embedding dimension was empirically chosen with respect to proposed in Refs. 67 and 68.

Entropy estimations in error bars with means and standard deviations are shown in Fig. 2. All three methods provided that patients had lower entropy at each scalp location in comparison to controls. In both patients and controls, the lowest entropy values were observed at parietal locations (P3, P4, Pz) and the highest entropy values were observed at temporal locations (T3, T4) among 19-channels. The performance of proposed features was evaluated by using SVM-based classification of patients and control groups. Each sample was represented by 19- and 8-dimensional feature vectors in entropy estimations and interhemispheric dependencies, respectively. In classification stage, we applied two fold cross-validation and partitioned the experimental data into two equal size subsets of samples randomly. We performed training on one subset and tested on the other. The training and classification were then repeated by changing the subsets. We iterated the 2-fold cross-validation process 20 times with different data folds for more robust statistical analysis to assess how the results generalize the unseen data.

The statistical analysis of classification results was performed using five different quantities. Sensitivity is true-positive rate (also called recall rate), which measures the proportion of correctly classified

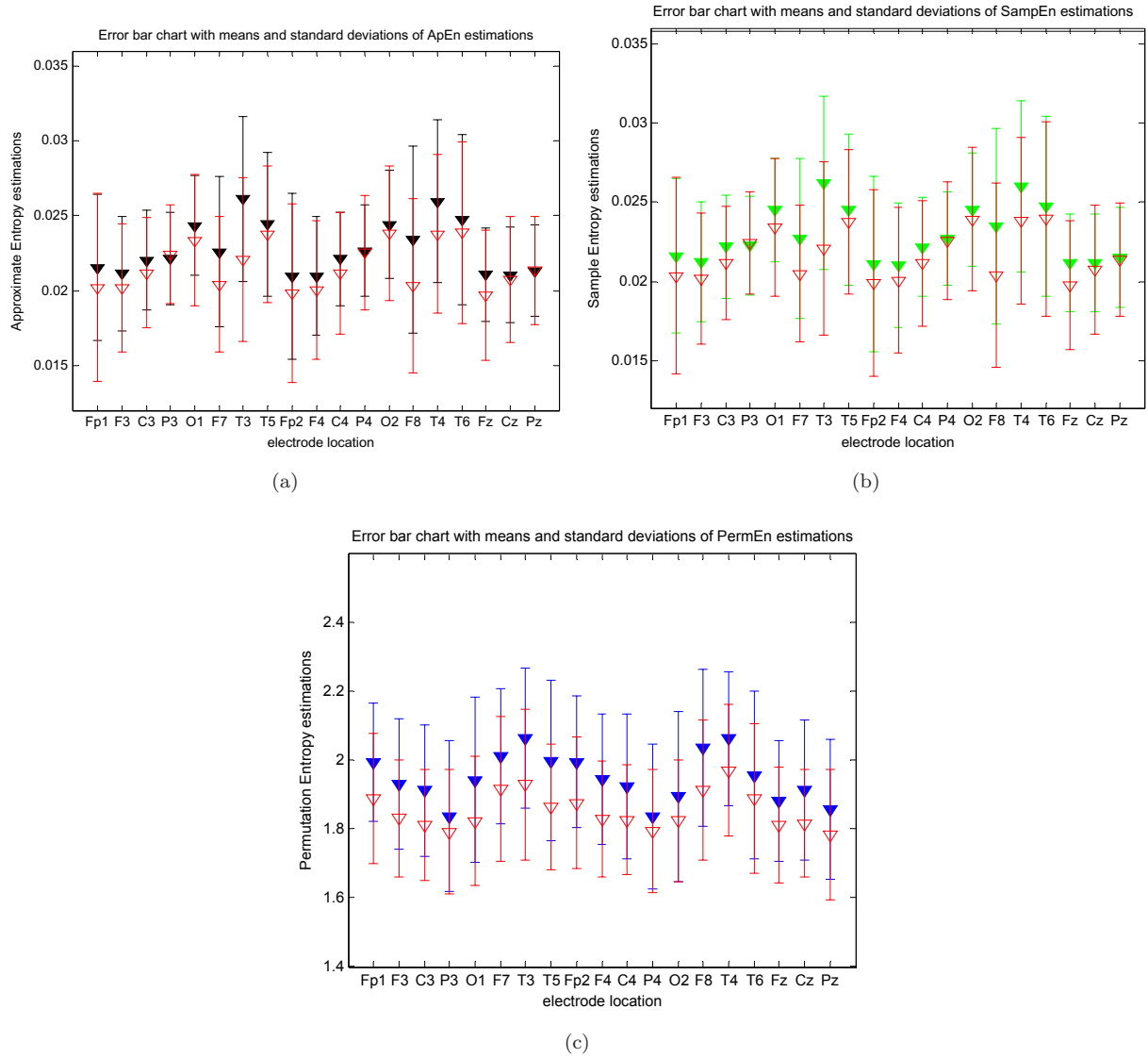


Fig. 2. (Color online) Error bars with means and stds of entropy estimations obtained by using (a) ApEn, (b) SampEn and (c) PermEn for patients (in black) and controls (in red).

OCD samples in all OCD samples. Specificity is true negative rate, which measures the proportion of correctly classified healthy controls in all healthy samples. Accuracy is the proportion of correctly classified sample (both OCD and normal) in total dataset. Precision is the proportion of correctly classified OCD samples in all samples classified as OCD.

The F -score is the harmonic mean of precision and sensitivity. The statistical analysis of classification results using the proposed features is given in Table 2 with those measurements. The results are shown by the mean and standard deviation of 20 trials. The clearest difference between patients and

controls can be detected by PermEn with respect to all performance criteria (sensitivity, specificity, precision, F -score and accuracy) in the common range of approximately 85%. Considering interhemispheric dependency estimations, the highest classification accuracy of 77.94% is obtained by using MI. Table 2 shows that all single channel complexity features provide the higher classification performance in comparison to two channel regional interhemispheric dependency features.

We also analyzed the results using receiver operating characteristic (ROC) curve, which plots the true positive rate against false positive rate.

Table 2. Statistical analysis of classification results.

Method	Statistical measures				
	Sensitivity	Specificity	Precision	<i>F</i> -score	Accuracy
ApEn	79.44 $\pm$ 5.4	80.22 $\pm$ 4.4	80.15 $\pm$ 3.7	0.80 $\pm$ 0.03	79.83 $\pm$ 3.5
SampEn	80.00 $\pm$ 5.9	79.44 $\pm$ 6.1	79.81 $\pm$ 4.6	0.80 $\pm$ 0.03	79.72 $\pm$ 3.6
PermEn	85.22 $\pm$ 7.4	84.78 $\pm$ 7.5	85.20 $\pm$ 6.3	0.85 $\pm$ 0.05	85.00 $\pm$ 5.2
Coherence	76.00 $\pm$ 9.3	75.00 $\pm$ 11.9	76.22 $\pm$ 6.3	0.76 $\pm$ 0.04	75.50 $\pm$ 4.0
MI	78.00 $\pm$ 10.0	77.89 $\pm$ 5.9	77.91 $\pm$ 5.4	0.78 $\pm$ 0.06	77.94 $\pm$ 5.7

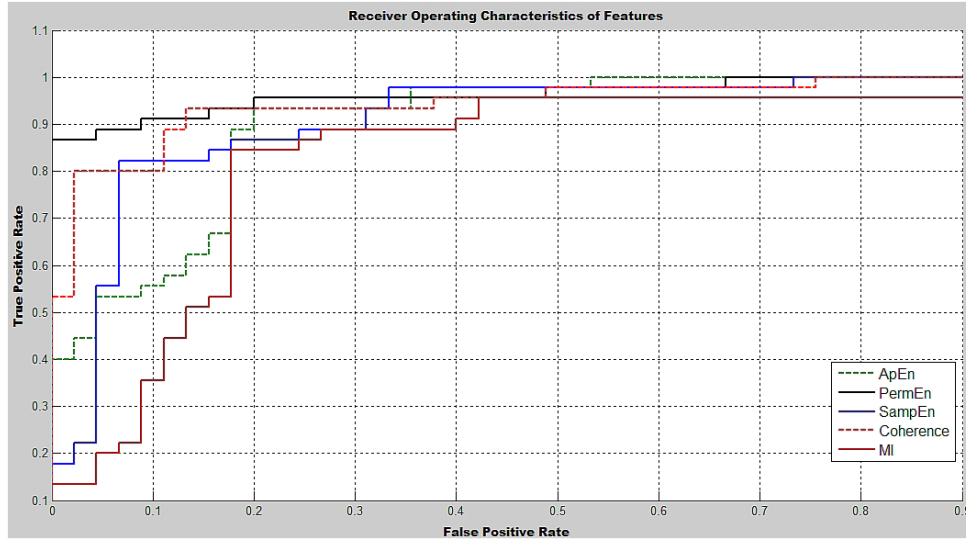


Fig. 3. ROC curves of classification results.

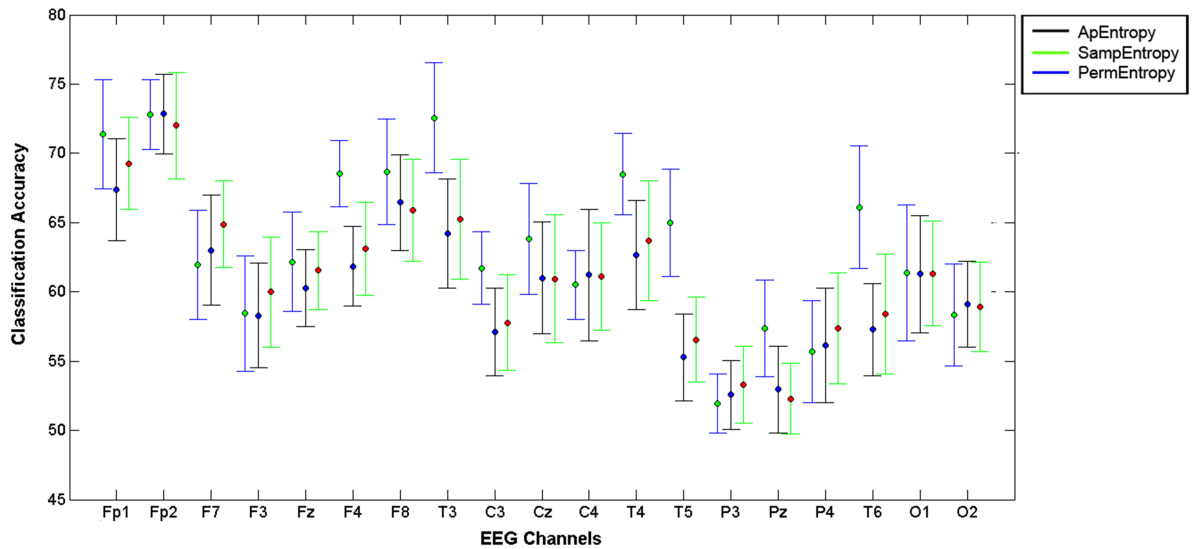


Fig. 4. Single channel classification results of entropy based features: classification accuracies with mean and standard deviation of 20 trials.

The curve depicts the classifier performance as the threshold used in SVM, i.e. the distance to the separating hyperplane, is varied. One of the interpretations of the plot is the area under ROC curve, called AUC, which is the portion of unit square area.

AUC value is always between 0 and 1.0, and gives the probability that the SVM assigns higher scores to a randomly chosen OCD sample than to a randomly chosen control sample. As the area gets larger, the classifier produces more accurate results.

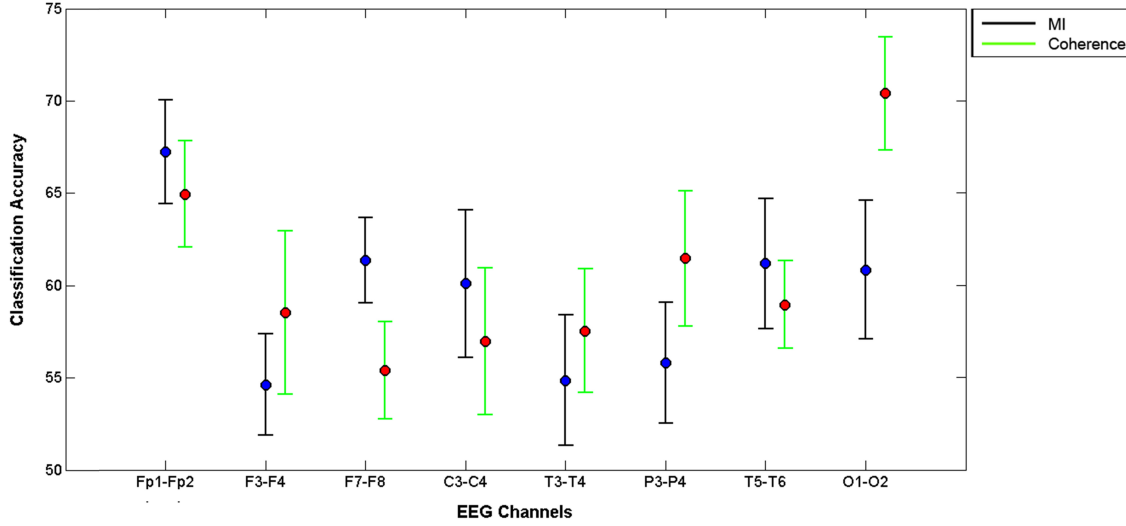


Fig. 5. Two channel classification results for synchrony-based features: classification accuracies with mean and standard deviation of 20 trials.

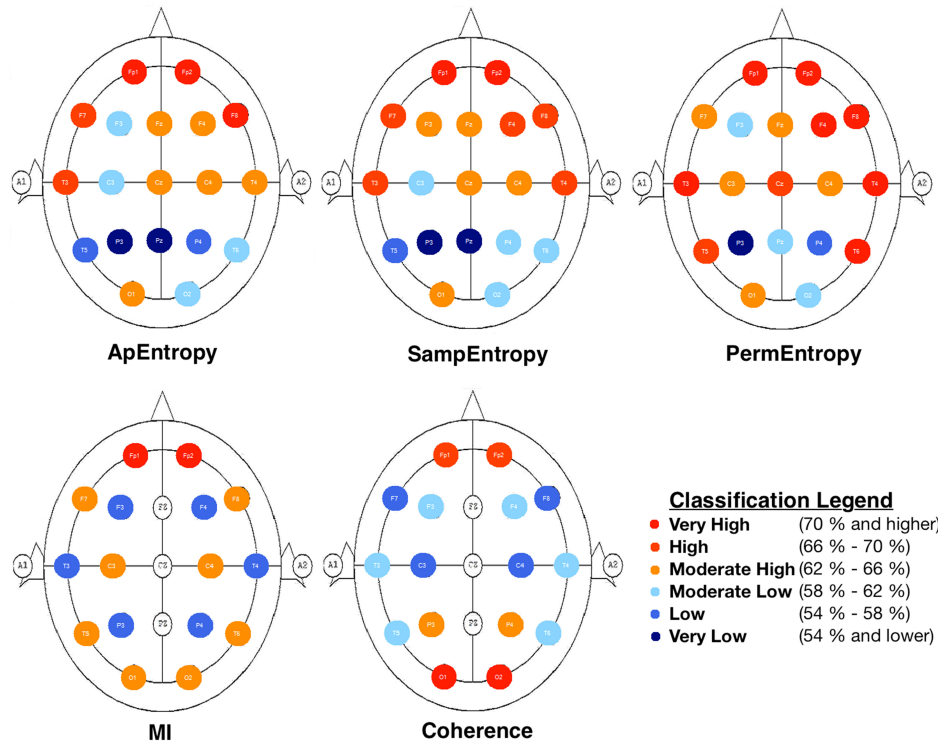


Fig. 6. Scalp plots of average classification accuracies computed for both single channel EEG complexity features and two channel interhemispheric dependency features as shown in Figs. 4 and 5, respectively.

The ROC curve analysis of all the features is given in Fig. 3. The highest AUC value is provided by single channel entropy values estimated by using PermEn whereas the lowest AUC value is originated from dependency features estimated by using MI. The classification performance is measured 20 times with different 2-fold cross-validation data folds for both single channel entropy-based EEG complexity features (19 features) and two channel regional interhemispheric dependency features (eight features) and then, classification accuracies in percent with mean and standard deviation of 20 trials are shown in Figs. 4 and 5, respectively. Regarding as Fig. 4, PermEn provides the accuracy higher than 70% at prefrontal (Fp1, Fp2), frontal (F4, F8) and temporal (T3, T4) locations, however, it provides the low accuracy lower than 60% at parietal locations (P3, P4, Pz). In particular, the best classification accuracy was originated from PermEn for pre-frontal features. Additionally, accuracy bars belong to PermEn for all temporal measurements (T3, T4, T5, T6) are clearly superior than that of both ApEn and SampEn. It also depicts that the accuracies increases, as the measurements moves from left to right lobe.

If we compare MI and coherence in itself regarding as Fig. 5, MI provides higher classification accuracies at both pre-frontal and frontal (F7, F8) locations. In particular, coherence shows the difference between patients with OCD and controls in occipital regions unlike our expectations.

The scalp plots of mean classification accuracies for both single channel EEG complexity features and two channel interhemispheric dependency features are also shown based on the international 10–20 electrode placement system in Fig. 6 to summarize all the experimental results. Coloring scale was determined with respect to classification accuracy in percent as stated in Fig. 6.

5. Discussion

In the present study, we examined single channel EEG complexity and two channel interhemispheric dependency measurements in classifying OCD and controls. By means of EEG complexity, the measurements in the right lobe are found to be more discriminative due to the decrease in EEG complexities of patients with OCD. Since, PermEn provides the higher classification accuracy ($\geq 70\%$) for

right frontal (F4, F8) and right temporal (T4, T6) measurements in comparison to left side measurements, we may say that the right hemisphere is more affected by OCD rather than the left lobe. The particular inference is that PermEn is superior to both ApEn and SampEn, since PermEn does not consider self-similarity between EEG segments and it converts all EEG segments on to a symbolic map prior to entropy estimation. However, all methods commonly provide that OCD is mainly characterized by the prefrontal dysfunctions as hypothesized in Ref. 50. In addition, two channel regional interhemispheric dependency measurements are less discriminative than the single channel complexity estimations. We think that this is because of the complexity differences between right and left lobe of OCD patients compared to controls. Probably, conventional coherence provides only spectral components in studying interhemispheric dependency. In Refs. 62, 71 and 72, wavelet coherence was found to be superior to conventional coherence in classifying Alzheimer's disease and controls over each EEG frequency band of it is due to high time resolution. In future work, we will calculate wavelet coherence over EEG sub-bands (delta (0–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz) and gamma (30–60 Hz)) to develop EEG-based diagnosis of OCD. EEG is the most important investigation in the diagnosis and management of many neuropsychiatric disorders, however, a loss in spatial resolution of EEG series collected from 19-electrode locations could be considered as disadvantage of EEG analysis. The use of brain scanning techniques such as conventional MRI and functional MRI (f-MRI) could provide three-dimensional noninvasive spatial mapping with high frequency resolution in observing neuronal activities in specific regions of the brain. In Refs. 74 and 75, functional connectivity analysis was studied on f-MRI data. Recent studies show that magnetoencephalographical signals can also be used to analyze functional connectivity of the brain to develop an efficient biomarker in diagnose.⁷⁶ In this study, functional connectivity graph, which combines graph theory and community pattern analysis as introduced in Ref. 78, was obtained from EEG series.⁷⁶

It is possible to design a future work consisting of simultaneously recording cortical EEG series and brain images to obtain more detailed analysis with high resolution in both time and frequency. In such

work, EEG characterization and MRI findings could be correlated after preprocessing of EEG to remove artifacts caused by the MRI environment. Nevertheless, the number of patients have to be more and more in this future study. As well, wide range of uncommon factors (the lack of consistency) in the population of participants could be considered as limit in the study.

It was presented in some clinical studies consisting of multichannel quantitative EEG analysis that OCD shows abnormalities at both frontal and temporal regions of the brain with respect to EEG band powers.^{3,51} Our recent results are consistent with these previous reports. The involvement of prefrontal cortex is consistent with research showing that OCD is an illness of orbitofronto-striatal structures.^{33,50,52}

6. Conclusion

Studies on other neuropsychiatric disorders have shown different patterns of complexity. An increased complexity has been observed in relatively younger patients with recent-onset schizophrenia with positive symptoms, whereas relatively older schizophrenics with negative symptoms and chronic illness exhibited a decreased complexity.⁵³ Clinical studies have demonstrated a higher complexity in several disorders such as depression,⁵⁴ major depressive disorder,⁵⁵ a manic episode of bipolar mood disorder⁵⁶ and attention deficit hyperactivity,⁵⁷ however, several studies have reported a lower complexity in other psychiatric disorders such as dissociative states,⁵⁸ anorexia nervosa⁵⁹ and panic disorder.⁶⁰ The last two, with which OCD was classified among anxiety disorders until recently, have the common characteristic of decreased complexity as in OCD.

Our finding that EEG differences are most obvious in the prefrontal areas is consistent with literature showing that primarily orbito-frontal cortex and basal ganglia is involved in OCD. Since the basal ganglia are deep in the brain and EEG records the cortical activity, striatal pathology may be overlooked by EEG and only frontocortical pathology can be apparent. Studies using functional MRI have founded a high degree of connectivity or hyperactivity in the orbito-frontal cortex and the basal ganglia in un-medicated OCD patients.^{63,64}

The fact that our patients were drug-naive is important in that suspicion about the modifying

effect of medications on brain function has already been relieved. EEG studies on schizophrenia⁶¹ and fMRI studies on OCD^{33,34} have commonly shown that pharmacotherapy leads to changes in brain activities. References 33 and 34 recruited OCD patients who had been un-medicated for some time, but not drug-naive. It cannot be excluded, however, that previous medication leads to changes in functional brain activities. Therefore, the present study reports clear electrophysiological pathology that is not confounded by pharmaco-medical interference.

Our small sample size is a shortcoming of the study. Comparisons of drug-naive patients with medicated and chronic patients, young and elder, and of pre- and post-treatment results with regard to complexity will be highly valuable in future studies. OCD is a disorder usually comorbid with a variety of other mental illnesses, and therefore the investigation of EEG data that differentiates co-occurring conditions may present interesting knowledge. Comparing the EEG data of patients having OCD with those suffering from other neuropsychiatric disorders, such as schizophrenia, will enhance our understanding of complexity in the central nervous system.

References

1. V. A. Arlington, *Diagnostic and Statistical Manual of Mental Disorders*, 5th edn. (American Psychiatric Association, 2013).
2. S. Kubota and K. Aihara, Computational model of obsessive compulsive disorder: A unified explanation of the treatments, *Int. J. Bifurcat. Chaos* **14**(1)(2004) 263–277.
3. F. Karadag, N. K. Oguzhanoglu, T. Kurt, A. Oguzhanoglu, F. Atesci and O. Ozdel, Quantitative EEG analysis in obsessive compulsive disorder, *Int. J. Neurosci.* **113** (2003) 833–847.
4. O. Pogarell, G. Juckel, P. Mavrogiorgou, C. Mulert, M. Folkerts, W. Hauke, M. Zaudig, H. J. Moller and U. Hegerl, Symptom-specific EEG power correlations in patients with obsessive-compulsive disorder, *Int. J. Psychophysiol.* **62** (2006) 87–92.
5. L. S. Prichep, F. Mas, E. Hollander, M. Liebowitz, E. R. John *et al.*, Quantitative electroencephalographic sub typing of obsessive-compulsive disorder, *Psychiatry Res.* **50** (1993) 25–32.
6. T. G. Bolwig, E. S. Hansen, A. Hansen, H. Merkin and L. S. Prichep, Toward a better understanding of the pathophysiology of OCD SSRI responders: QEEG source localization, *Acta Psychiatr. Scand.* **115** (2007) 237–242.
7. L. F. Fontenelle, M. V. Mendlowicz, P. Ribeiro, R. A. Piedade and M. Versiani, Low-resolution

- electromagnetic tomography and treatment response in obsessive-compulsive disorder, *Int. J. Neuropsychopharmacol.* **9** (2006) 89–94.
8. H. Adeli and S. G. Dastidar, *Automated EEG-Based Diagnosis of Neurological Disorders: Inventing the Future of Neurology* (CRC Press, Taylor & Francis, Boca Raton, Florida, 2010).
9. U. R. Acharya, F. Molinari, S. V. Sree, S. Chattopadhyay, K. Hoong and J. S. Suri, Automated diagnosis of epileptic EEG using entropies, *Biomed. Signal Proc. Control* **7** (2012) 401–408.
10. U. R. Acharya, S. V. Sree, S. Chattopadhyay and J. S. Suri, Automated diagnosis of normal and alcoholic EEG signals, *Int. J. Neural Syst.* **22**(3) (2012) 1250011.
11. R. J. Martis, U. R. Acharya, J. H. Tan, A. Petznick *et al.*, Application of empirical mode decomposition for automated detection of epilepsy using EEG signals, *Int. J. Neural Syst.* **22**(6) (2012) 1250027.
12. J. S. Richman and J. R. Moorman, Physiological time series analysis using approximate and sample entropy, *Am. J. Physiol.: Heart Circ. Physiol.* **278**(6) (2000) H2039–49.
13. D. Abásolo, R. Hornero, P. Espino, J. Poza, C. I. Sánchez and R. de la Rosa, Analysis of regularity in the EEG background activity of Alzheimer's disease patients with approximate entropy, *Clin. Neurophysiol.* **116**(8) (2005) 1826–1834.
14. S. Aydın, H. M. Saraoğlu and S. Kara, Log energy entropy based EEG classified with multilayer neural networks in seizure, *Ann. Biomed. Eng.* **37**(12) (2009) 2626–2630.
15. S. M. Pincus and A. L. Goldberger, Physiological time-series analysis: What does regularity quantify? *Am. J. Physiol.: Heart Circ. Physiol.* **266**(4) (1994) H1643–56.
16. R. Ferenets, T. Lipping, A. Anier, *et al.*, Comparison of entropy and complexity measures for the assessment of depth of sedation, *IEEE Trans. Biomed. Eng.* **53**(6) (2006) 1067.
17. D. E. Lake, J. S. Richman, M. P. Griffin and J. R. Moorman, Sample entropy analysis of neonatal heart rate variability, *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **283**(3) (2002) R789–97.
18. Y. Song and P. Lio, A new approach for epileptic seizure detection: Sample entropy based feature extraction and extreme learning machine, *J. Biomed. Sci. Eng.* **3** (2010) 556–567.
19. Y. Wang, Z. Liang, L. J. Voss, J. W. Sleight and X. Li, Multiscale sample entropy of EEG during sevoflurane anesthesia, *J. Clin. Monit. Comput.* **28**(4) (2014) 409–417.
20. J. M. Amigo, M. B. Kennel and L. Kocarev, The permutation entropy rate equals the metric entropy rate for ergodic information sources and ergodic dynamical systems, *Physica D* **210** (2005) 77–95.
21. C. Bandt and B. Pompe, Permutation entropy: A natural complexity measure for time series, *Phys. Rev. Lett.* **88** (2002) 174102.
22. A. A. Bruzzo, B. Gesierich, M. Santi, C. A. Tassinari, N. Birbaumer and G. Rubboli, Permutation entropy to detect vigilance changes and preictal states from scalp EEG in epileptic patients. A preliminary study, *Neurol. Sci.* **29**(1) (2008) 3–9.
23. Y. Cao, W. W. Tung, J. B. Gao, V. A. Protopopescu and L. M. Hively, Detecting dynamical changes in time series using the permutation entropy, *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* **70** (4 Pt 2) (2004) 046217.
24. M. Sabeti, Entropy and complexity measures for EEG signal classification of schizophrenic and control participants, *Artif. Intell. Med.* **47**(3) (2009) 263–274.
25. D. Abasolo, R. Hornero and P. Espino, Approximate entropy of EEG background activity in Alzheimer's disease patients, *Intell. Autom. Soft Comput.* **15**(4) (2009) 591–603.
26. L. Wentian, Mutual information functions versus correlation functions, *J. Stat. Phys.* **60**(5–6) (1990) 823–837.
27. J. Xu, Z. R. Liu, R. Liu and Q. F. Yang, Information transmission in human cerebral cortex, *Physica D* **106** (1997) 363–374.
28. S. H. Na, S. H. Jin, S. Y. Kim and B. J. Ham, EEG in schizophrenic patients: Mutual information analysis, *Clin. Neurophysiol.* **113**(12) (2002) 1954–1960.
29. J. Jeonga, J. C. Goreb and B. S. Petersona, Mutual information analysis of the EEG in patients with Alzheimer's disease, *Clin. Neurophysiol.* **112** (2001) 827–835.
30. G. Nolte, O. Bai, L. Wheaton, Z. Mari, S. Vorbach and M. Hallet, Identifying true brain interaction from EEG data using the imaginary part of coherency, *Clin. Neurophysiol.* **115** (2004) 2292–2307.
31. D. Lehmann, W. K. Strik, B. Henggeler, T. Koenig and M. Koukkou, Brain electric microstates and momentary conscious mind states as building blocks of spontaneous thinking: I. Visual imagery and abstract thoughts, *Int. J. Psychophysiol.* **29**(1) (1998) 1–11.
32. S. T. Okuhata, S. Okazaki and H. Maekawa, EEG coherence pattern during simultaneous and successive processing task, *Int. J. Psychophysiol.* **72** (2009) 89–96.
33. J. C. Beucke, J. Sepulcre, T. Talukdar, C. Linnman, K. Zschenderlein, T. Endrass *et al.*, Abnormally high degree connectivity of the orbitofrontal cortex in obsessive-compulsive disorder, *JAMA Psychiatry* **70** (2013) 619–629.
34. T. Nakao, A. Nakagawa, T. Yoshiura, E. Nakatani, M. Nabeyama, C. Yoshizato *et al.*, Brain activation of patients with obsessive-compulsive disorder

- during neuropsychological and symptom provocation tasks before and after symptom improvement: A functional magnetic resonance imaging study, *Biol. Psychiatry* **57** (2005) 901–910.
35. C. F. V. Latchoumane, D. Chung, S. Kim and J. Jeong, Segmentation and characterization of EEG during mental tasks using dynamical non stationarity, in *Proc. Computational Intelligence in Medical and Healthcare* (Plymouth, England, 2007), pp. 1–5.
36. M. T. Davis and O. Roger, *Disorders of Personality: DSM-IV and Beyond*, 2nd edn. (John Wiley & Sons, Oxford, England, 1996), pp. xiii 818.
37. W. K. Goodman, L. H. Price et al., The Yale-Brown obsessive compulsive scale II. Validity, *Arch. Gen. Psychiatry* **46**(11) (1989) 1012–1016.
38. J. B. W. Williams, A structured interview guide for the Hamilton depression rating scale, *Arch. Gen. Psychiatry* **45**(8) (1988) 742–747.
39. A. T. Beck, E. N. Brown and G. S. A. Robert, An inventory for measuring clinical anxiety: Psychometric properties, *J. Consult. Clin. Psychol.* **56**(6) (1988) 893–897.
40. A. A. Sergejew and A. C. Tsoi, Markovian analysis of EEG signal dynamics in obsessive-compulsive disorder, in *Advances in Processing and Pattern Analysis of Biological Signals*, eds. I. Gath and G. F. Inbaa (Plenum Press, New York, 1995).
41. T. Schreiber, Detecting and analyzing non stationarity in a time series using nonlinear cross predictions, *Phys. Rev. Lett.* **78**(5) (1997) 843–846.
42. J. B. Gao, Detecting non stationarity and state transitions in a time series, *Phys. Rev. E. Stat. Nonlin. Soft Matter Phys.* **63**(6 Pt 2) (2001) 066202.
43. J. Wackermann, D. Lehmann, C. M. Michel and W. K. Strik, Adaptive segmentation of spontaneous EEG map series into spatially defined microstates, *Int. J. Psychophysiol.* **14**(3) (1993) 269–283.
44. E. Pereda, Q. R. Quiroga and J. Bhattacharya, Non-linear multivariate analysis of neurophysiological signals, *Prog. Neurobiol.* **77** (2005) 1–37.
45. Z. Sankari, H. Adeli and A. Adeli, Intrahemispheric, interhemispheric and distal EEG coherence in Alzheimer's disease, *Clin. Neurophysiol.* **122** (2011) 897–906.
46. L. C. Lin, C. S. Ouyang, C. T. Chiang, R. C. Yang, R. C. Wu and H. C. Wu, Early prediction of medication refractoriness in children with idiopathic epilepsy based on scalp EEG analysis, *Int. J. Neural Syst.* **24**(7) (2014) 1450023.
47. Y. Wang, W. Zhou, Q. Yuan and X. Li, Comparison of fractal features of ictal and interictal EEGs, *Int. J. Neural Syst.* **23**(6) (2013) 1350028.
48. V. Jumutc, P. Zayakin and A. Borisov, Ranking-based kernels in applied biomedical diagnostics using a support vector machine, *Int. J. Neural Syst.* **21**(6) (2011) 459–473.
49. V. Vapnik, *Statistical Learning Theory* (Wiley Chichester, GB, 1998).
50. P. Cavedini, S. Ferri, S. Scarone and L. Bellodi, Frontal lobe dysfunction in obsessive-compulsive disorder and major depression: A clinical neuropsychological study, *Psychiatry Res.* **78** (1998) 21–28.
51. O. Pogarell, G. Juckel, P. Mavrogiorgou, C. Mulert, M. Folkerts et al., Symptom-specific EEG power correlations in patients with obsessive compulsive disorder, *Int. J. Psychophysiol.* **62** (2006) 87–92.
52. L. Menzies, S. R. Chamberlain, A. R. Laird, S. M. Thelen, B. J. Sahakian and E. T. Bullmore, Integrating evidence from neuroimaging and neuropsychological studies of obsessive-compulsive disorder: The orbitofronto-striatal model revisited, *Neurosci. Biobehav. Rev.* **32**(3) (2008) 525–549.
53. R. Fernandez, J. J. Gomez and L. Hornero, Complexity and schizophrenia, *Prog. Neuropsychopharmacol. Biol. Psychiatry* **45** (2013) 267–276.
54. M. Méndez, P. Zuluaga, R. Hornero, C. Gómez, J. Escudero, A. R. Palancas et al., Complexity analysis of spontaneous brain activity: Effects of depression and antidepressant treatment, *J. Psychopharmacol.* **26** (2012) 636–664.
55. M. Ahmadi, H. Adeli and A. Adeli, Fractality analysis of frontal brain in major depressive disorder, *Int. J. Psychophysiol.* **85** (2012) 206–211.
56. B. Bahrami, R. Seyedsadjadi, B. Babadi and M. Noroozian, Brain complexity increases in mania, *Neuroreport* **16**(2) (2005) 187.
57. A. Fernandez, J. Quintero, R. Hornero et al., Complexity analysis of spontaneous brain activity in attention deficit hyper activity disorder: Diagnostic implication, *Biol. Psychiatry* **65** (2009) 571–577.
58. P. Bob and M. Svetlak, Dissociative states and neural complexity, *Brain Cogn.* **75**(2) (2011) 188–195.
59. E. Toth, I. Kondakor and F. Tury, Nonlinear and linear EEG complexity changes caused by gustatory stimuli in anorexia nervosa, *Int. J. Psychophysiol.* **51**(3) (2004) 253–260.
60. P. Bob, M. Kukleta, I. Riečanský and M. Susta, Chaotic EEG patterns during recall of stressful memory related to panic attack, *Physiol. Res.* **55** (2006) S113–S119.
61. T. Takahashi, R. Y. Cho, T. Mizuno, M. Kikuchi, T. Murata, K. Takahashi et al., Antipsychotics reverse abnormal EEG complexity in drug naive schizophrenia: A multiscale entropy analysis, *Neuroimage* **51** (2010) 173–182.
62. Z. Sankari, H. Adeli and A. Adeli, Wavelet coherence model for diagnosis of Alzheimer's disease, *Clin. EEG Neurosci.* **43** (2012) 268–278.
63. D. H. Kang, J. H. Jang, J. Y. Han, J. H. Kim, W. H. Jung, J. S. Choi et al., Neural correlates of altered response inhibition and dysfunctional connectivity at rest in obsessive compulsive disorder,

- Prog. Neuro-Psychopharmacol. Biol. Psychiatry* **40** (2013) 340–346.
64. Y. Sakai, J. Narumoto, S. Nishida, T. Nakamae, K. Yamada, T. Nishimura *et al.*, Cortico striatal functional connectivity in non-medicated patients with obsessive-compulsive disorder, *Eur. Psychiatry* **26**(7) (2011) 463–469.
65. J. P. Eckmann and D. Ruelle, Ergodic theory of chaos and strange attractors, *Rev. Mod. Phys.* **57**(3) (1985) 617–656.
66. F. Takens, Mechanical and gradient systems: Local perturbations and generic properties, *Bol. Soc. Brasil. Mat.* **14**(2) (1983) 47–162.
67. G. Manis, Fast computation of approximate entropy, *Comput. Methods Prog. Biomed.* **91** (2008) 48–54.
68. Y. H. Pan, Y. H. Wang, S. F. Liang and K. T. Lee, Fast computation of sample entropy and approximate entropy in biomedicine, *Comput. Methods Prog. Biomed.* **104** (2011) 382–396.
69. M. Ahmadlou, H. Adeli and A. Adeli, Fractality and a wavelet chaos neural network methodology for EEG based diagnosis of autistic spectrum disorder, *J. Clin. Neurophysiol.* **27**(5) (2010) 328–333.
70. M. Ahmadlou, H. Adeli and A. Adeli, Fractality and a wavelet chaos methodology for EEG based diagnosis of Alzheimer's disease, *Alzheimer Dis. Assoc. Disord.* **25**(1) (2011) 85–92.
71. H. Adeli, S. G. Dastidar and N. Dadmehr, A spatio-temporal wavelet-chaos methodology for EEG based diagnosis of Alzheimer's disease, *Neurosci. Lett.* **444**(2) (2008) 190–194.
72. M. Ahmadlou and H. Adeli, Wavelet synchronization methodology: A new approach for EEG based diagnosis of ADHD, *Clin. EEG Neurosci.* **41**(1) (2010) 1–10.
73. D. Li, L. Xu, E. Goodman, Y. Xu and Y. Wu, Integrating a statistical background-foreground extraction algorithm and SVM classifier for pedestrian detection and tracking, *Integr. Comput. Aided Eng.* **20**(3) (2013) 201–216.
74. D. Rangaprakash, X. Hu and G. Deshpande, Phase synchronization in brain networks derived from correlation between probabilities of recurrences in functional MRI data, *Int. J. Neural Syst.* **23**(2) (2013) 1350003.
75. P. Piaggi, D. Menicucci, C. Gentili, G. Handjaras, A. Gemignani and A. Landi, Singular spectrum analysis and adaptive filtering enhance the functional connectivity analysis of resting state FMRI data, *Int. J. Neural Syst.* **24**(3) (2014) 1450010.
76. M. Ahmadlou, A. Adeli, R. Bajo and H. Adeli, Complexity of functional connectivity networks in mild cognitive impairment patients during a working memory task, *Clin. Neurophysiol.* **125**(4) (2014) 694–702.
77. Y. Wang, W. Zhou, Q. Yuan, X. Li, Q. Meng, X. Zhao and J. Wang, Comparison of ictal and interictal EEG signals using fractal features, *Int. J. Neural Syst.* **23**(6) (2013) 1350008.
78. M. Ahmadlou and H. Adeli, Functional community analysis of brain: A new approach for EEG based investigation of the brain pathology, *Neuroimage* **58** (2011) 401–408.
79. Q. Yuan, W. Zhou, S. Yuan, X. Li, J. Wang and G. Jia, Epileptic EEG classification based on kernel sparse representation, *Int. J. Neural Syst.* **24**(4) (2014) 1450015.
80. H. Adeli, S. Ghosh-Dastidar and N. Dadmehr, A wavelet-chaos methodology for analysis of EEGs and EEG subbands to detect seizure and epilepsy, *IEEE Trans. Biomed. Eng.* **54**(2) (2007) 205–211.
81. S. Ghosh-Dastidar, H. Adeli and N. Dadmehr, Mixed-band wavelet-chaos-neural network methodology for epilepsy and epileptic seizure detection, *IEEE Trans. Biomed. Eng.* **54**(9) (2007) 1545–1551.
82. M. Ahmadlou, H. Adeli and A. Adeli, New diagnostic EEG markers of the Alzheimer's disease using visibility graph, *J. Neural Transm.* **117**(9) (2010) 1099–1109.
83. Z. Sankari and H. Adeli, Probabilistic neural networks for EEG-based diagnosis of Alzheimer's disease using conventional and wavelet coherence, *J. Neurosci. Methods* **197**(1) (2011) 165–170.
84. M. Ahmadlou and H. Adeli, Fuzzy synchronization likelihood with application to attention-deficit/hyperactivity disorder, *Clin. EEG Neurosci.* **42**(1) (2011) 6–13.
85. M. Ahmadlou, H. Adeli and A. Adeli, Graph theoretical analysis of organization of functional brain networks in ADHD, *Clin. EEG Neurosci.* **43**(1) (2012) 5–13.
86. M. Ahmadlou, H. Adeli and A. Adeli, Improved visibility graph fractality with application for diagnosis of Autism spectrum disorder, *Physica A* **391**(20) (2012) 4720–4726.
87. M. Ahmadlou, H. Adeli and A. Adeli, Fuzzy synchronization likelihood-wavelet methodology for diagnosis of Autism spectrum disorder, *J. Neurosci. Methods* **211**(2) (2012) 203–209.
88. M. Kwon, S. Kavuri and M. Lee, Action-perception cycle learning for incremental emotion recognition in a movie clip using 3D fuzzy GIST based on visual and EEG signals, *Integr. Comput. Aided Eng.* **21**(3) (2014) 295–310.
89. L. J. Herrera, C. Fernandes, A. M. Mora, D. Migotina, R. Largo, A. Guillén and A. C. Rosa, Combination of heterogeneous EEG feature extraction methods and stacked sequential learning for sleep stage classification, *Int. J. Neural Syst.* **23**(3) (2014) 1350012.
90. G. R. Bermudeza, P. J. Garcia-Laencina and J. Roca-Dorda, Efficient automatic selection and

- combination of EEG features in least squares classifiers for motor-imagery brain computer interfaces, *Int. J. Neural Syst.* **23**(4) (2013) 1350015.
91. A. Temko, G. Boylan, W. Marnane and G. Lightbod, Robust neonatal EEG classification through adaptive background modeling, *Int. J. Neural Syst.* **23**(4) (2013) 1350018.
92. M. Ahmadlou and H. Adeli, Enhanced probabilistic neural network with local decision circles: A robust classifier, *Integr. Comput. Aided Eng.* **17**(3) (2010) 197–210.